

TOPIC 6: SEX DETERMINATION SYSTEMS & HONEYBEE GENETICS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: _____

Roll / Reg. No: _____

Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

1. Which chromosome mechanism of sex determination is found in domestic fowls and birds?
(A) XX – XY (B) XX – XO
(C) ZZ – ZW (Female heterogamety) (D) Haplodiploidy
2. In grasshoppers and bugs, sex determination operates via:
(A) XX female and XO male (B) ZW female and ZZ male
(C) XY female and XX male (D) Environmental temperature
3. The nuclear structure called 'X-body' was first observed during insect spermatogenesis by:
(A) Walter Sutton (1902) (B) Hermann Henking (1891)
(C) Thomas Morgan (1910) (D) Langdon Down (1866)
4. In honeybees, fertile queen and worker females are diploid (2n) with how many chromosomes?
(A) 16 chromosomes (B) 32 chromosomes
(C) 46 chromosomes (D) 24 chromosomes

SECTION B: Fill in the Blanks (1 Mark Each)

5. In XX-XY and XX-XO systems, the sex producing two distinct types of gametes is called male
6. In honeybees, unfertilized haploid eggs develop into male drones through the process of
7. Male honeybees produce sperms through the process of division.
8. Chromosomes not involved in sex determination are collectively called

SECTION C: Very Short Answer Questions (2 Marks Each)

9. [2M] Differentiate between Male Heterogamety and Female Heterogamety with examples.

10. [2M] What are Allosomes and Autosomes? Give the human chromosome counts for each.

SECTION D: Short Answer Questions (3 Marks Each)

11. [3M] Explain XY type of sex determination in humans with a schematic cross. Why is the male responsible for child gender?

12. [3M] Explain Haplodiploid sex determination in honeybees (*Apis mellifera*).

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. [5M] Explain sex determination mechanisms across animal kingdom: (a) Male heterogamety (XX-XY, XX-XO) (b) Female heterogamety (ZZ-ZW) with schematic diagrams.

14. [5M] Describe haplodiploid sex determination in honeybees. Justify why male drones have no father and cannot have sons, but have grandfathers and grandsons.